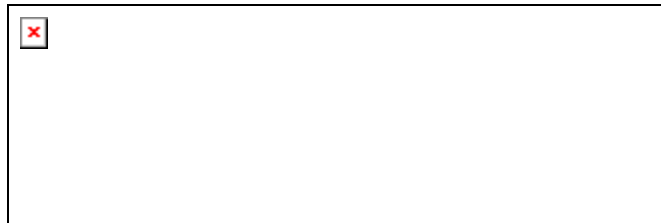




Introduction to Thai National Grid Project



**Thai National Grid Center
Software Industry Promotion Agency (Public Organization)
Ministry of Information and Communication Technology**



2006/12/14

<http://www.thaigrid.net>

What is Grid Computing Technology?

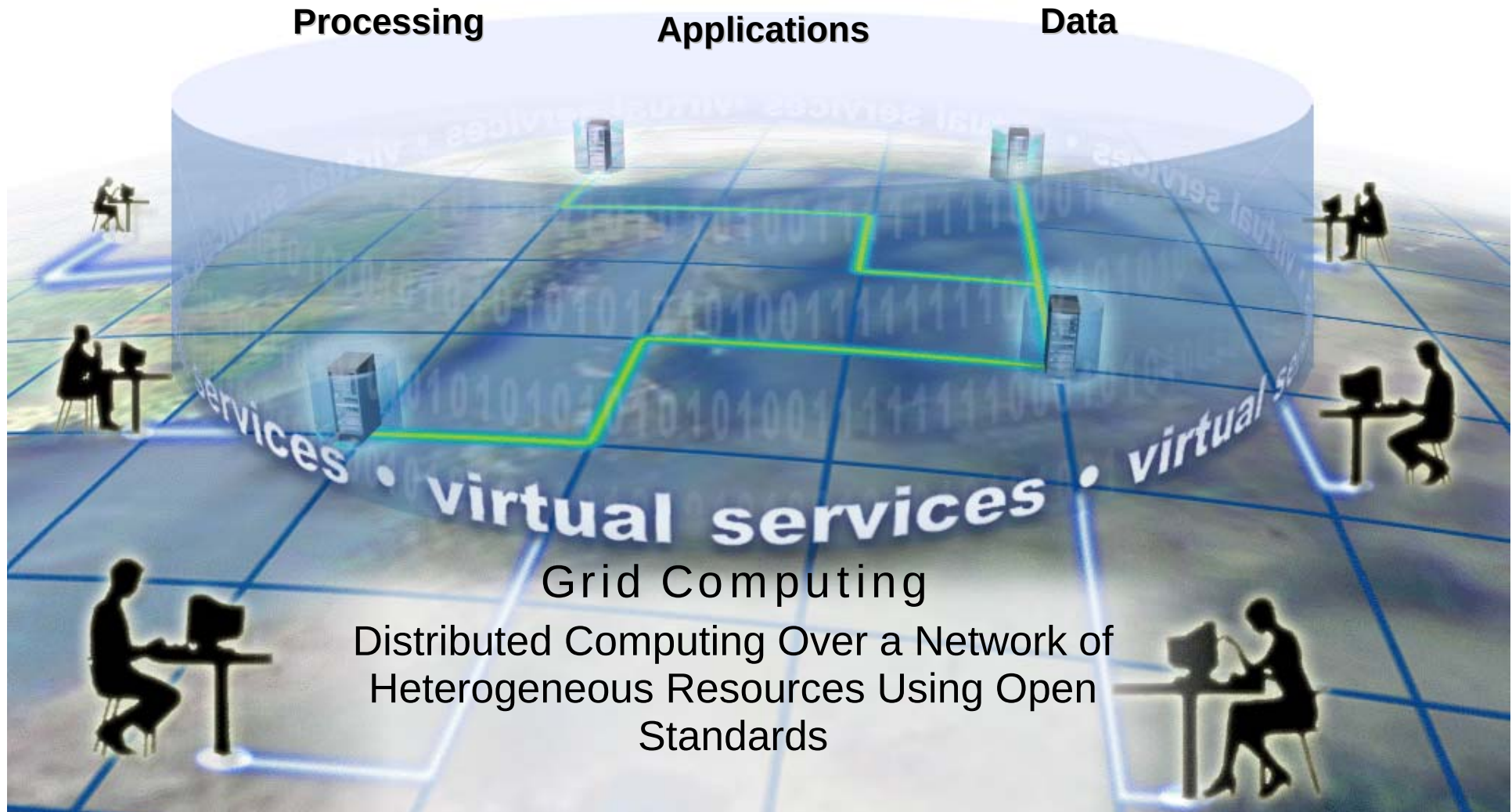


I/O Storage Operating System

Processing

Applications

Data



Grid Computing

Distributed Computing Over a Network of
Heterogeneous Resources Using Open
Standards

Impact of Grid Computing Technology

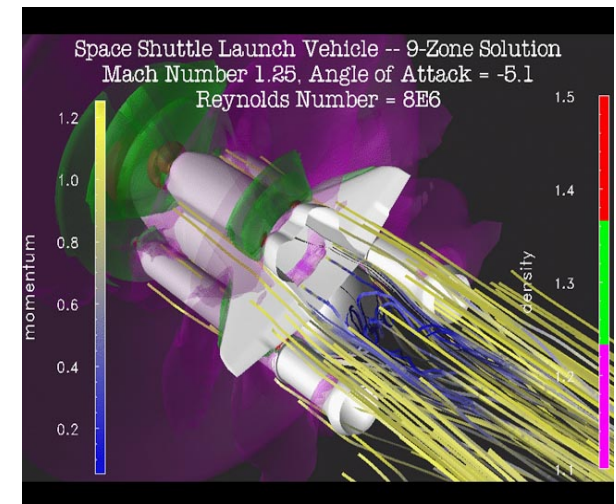


- What grid computing can do?
 - Efficient resources sharing
 - Provide a massive computing power
 - Utilize under used IT resources
 - Secure data sharing



Potential Application Domain

- Using Grid as Enterprise Computing Infrastructure
 - E-commerce
 - E-government
 - Health care Grid
 - E-learning
- Using Grid Computing Power for Industry
 - Digital Media and Animation
 - Computer Aided Design and Engineering
 - Industrial Simulation
 - Financial Analysis
 - Knowledge Discovery, Data Mining



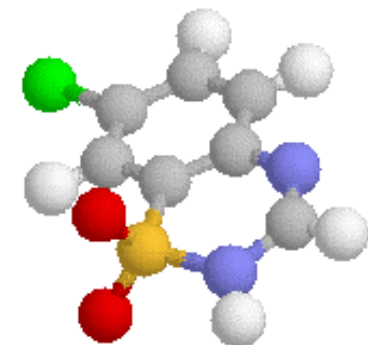
Potential Application Domain

- Using Grid Computing Power and Storage to for Research and Education Competitiveness
 - Life-science Grid
 - Cheminformatics Grid
 - Geoinformatics Grid
 - Digital Archive and E-learning
 - Computational Fluid Dynamics Grid , FEM
 - Simulation and Modeling Grid
 - Biomedical Imaging



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Benefits to Thailand



- Strengthen national competitiveness
 - Create infrastructure for key strategic research areas: Biotech, Nanotech, Life science (Avian Flu)
 - Support science and technology man power creation (Master, Ph.D)
 - Attract/keep leading computational researchers
- Strengthen long term IT strategies
 - e-commerce, e-government, e-learning
- Large cluster promotes science and engineering excellence
 - Allows all computational scientists/engineers to scale research into new regimes
 - Allows cluster and grid research to scale to large systems
- Create a new push for open source and Linux technology



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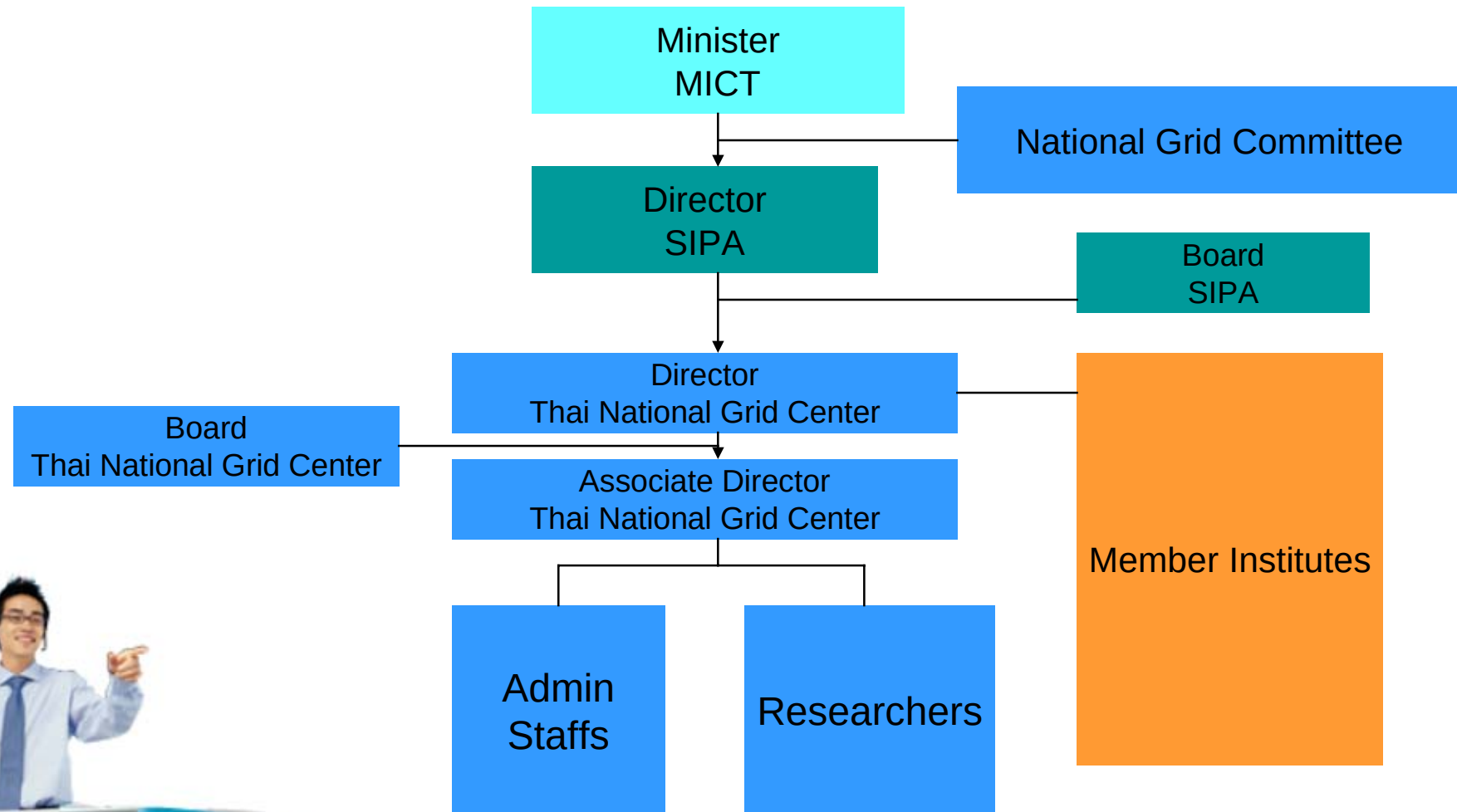
Thai National Grid Project



- A 3 years project under Ministry of Information and Communication Technology
- Start from 13 universities and 1 government organizations
- Goal
 - Building the next generation advanced computing infrastructure
 - Stimulating the adoption of grid technology to support research, education, and industry
 - Building up man power for future generation IT industry



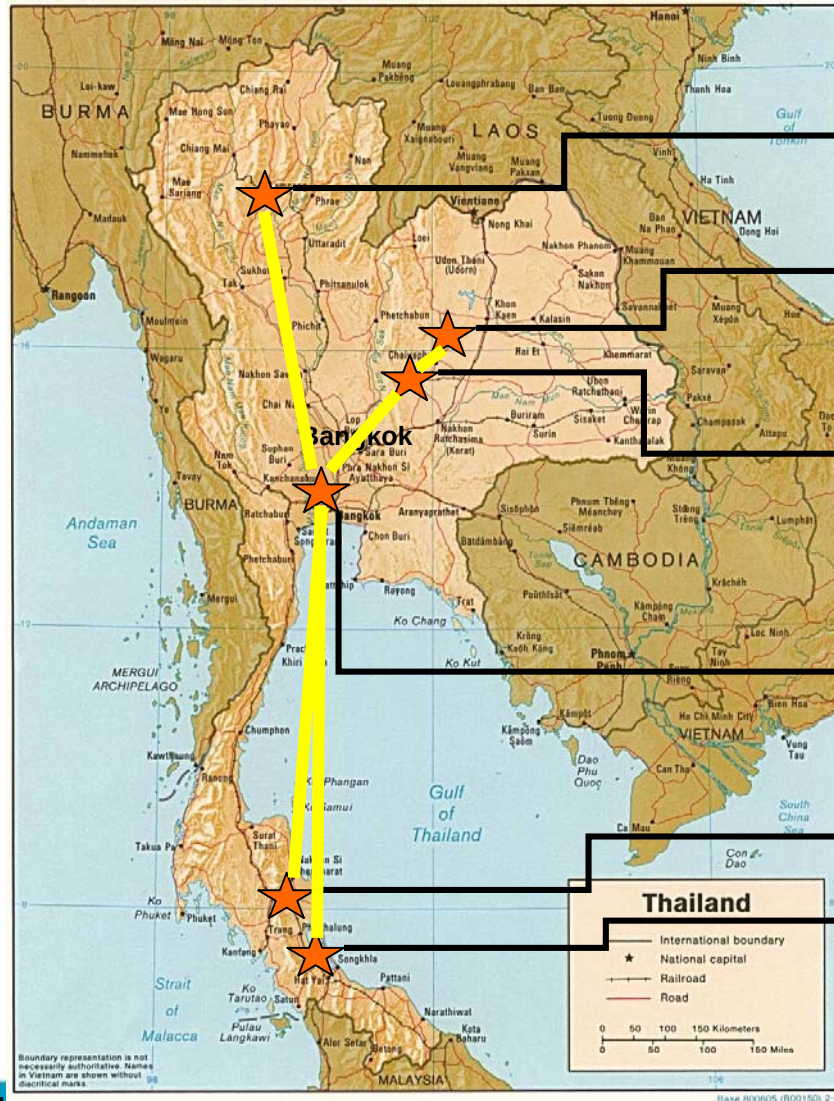
Project Organization



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Partners Institutes



Chaingmai University (CMU)

Khonkhan University (KKU)

Suranaree University of Technology (SUT)

Chulalongkorn University (CU)

Kasetsart University (KU)

King Mongkut's Institute of Technology North Bangkok (KMITNB)

King Mongkut's University of Technology Thonburi (KMUTT)

King Mongkut's Institute of Technology Ladkrabang (KMITL)

Mahidol University (MU)

Silpakorn University (SU)

The Meteorological Department Of Thailand (TMD)

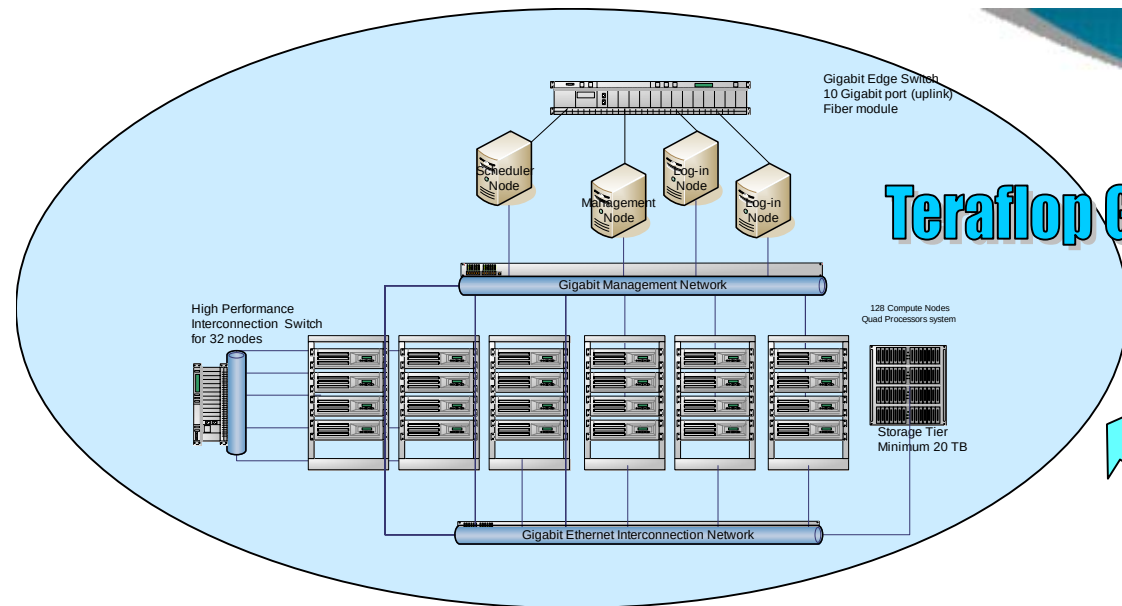
Walailak University (WU)

Prince of Songkla University (PSU)

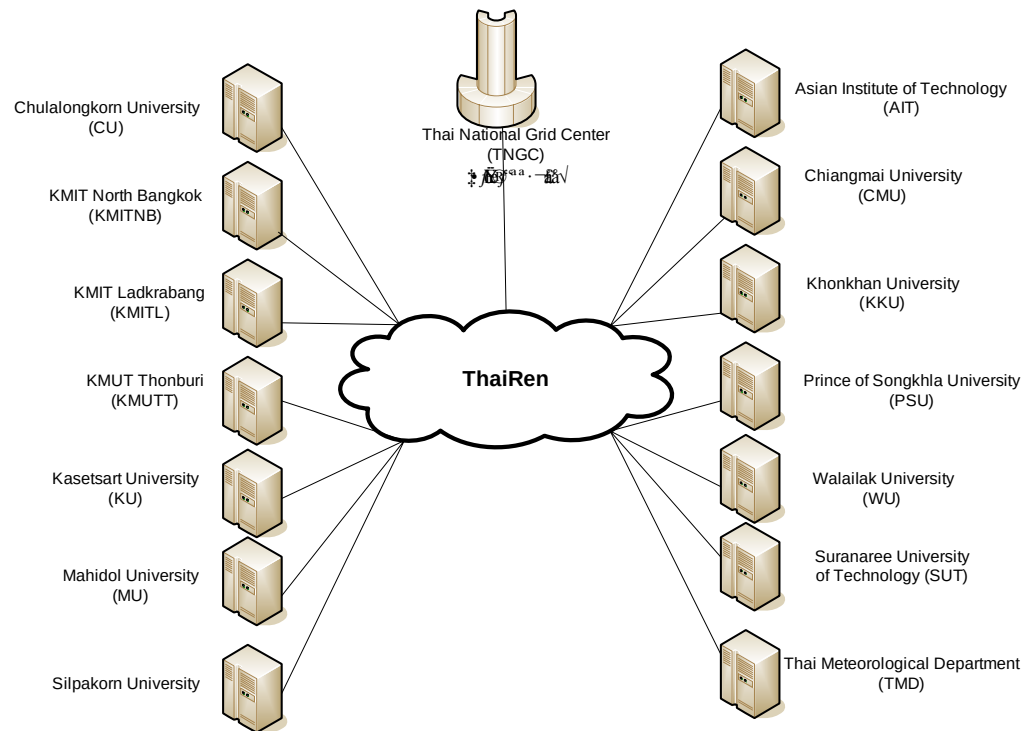


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Teraflop Grid Server



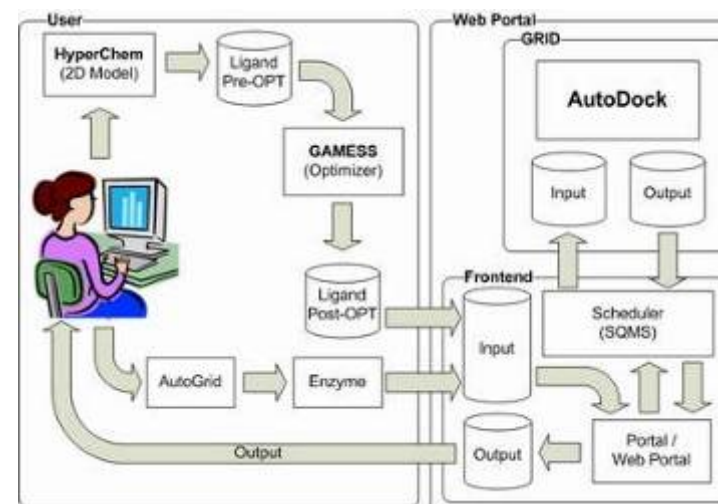
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ThaiGrid Drug Discovery Infrastructure



- Partners:
 - Dr. Jak Saengma, KU
 - Somsak Sriprayoonsakul, TNGC
 - IBM Thailand
- Challenges
 - From over 10000 active compounds available from Thai medicinal plants database, find a smallest set of compounds that has a potential to be used as a drug
 - Very compute intensive. Several month of computing time
- Solution: Use grid to increase computing power to 10-100 times
 - Distributed screening process to hundred of hosts on the Grid
- Benefit
 - Speeding up time to results
 - Allow scientists to spend more time analyzing results



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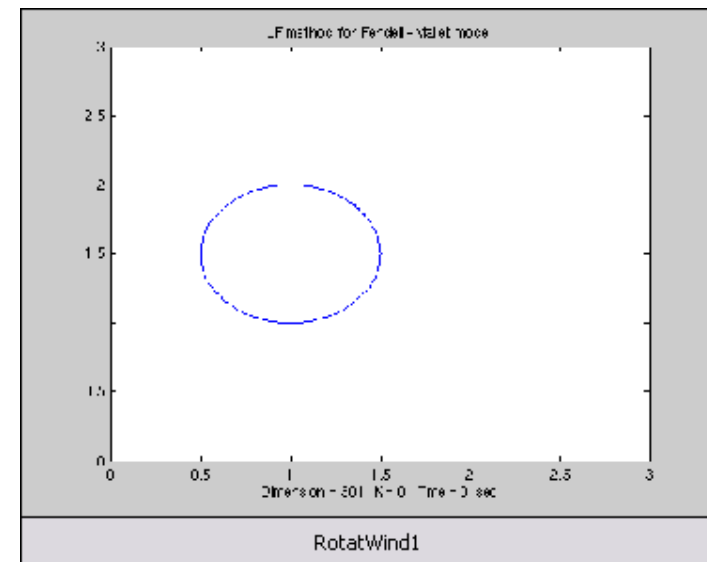
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Implicit Narrowband Level Set Method for Fire Spread Model



Dr. Pallop Huabsomboon
Mahidol University

- Challenges
 - Simulation of large scale firespread for forest fire
 - Very time consuming process
- Solution
 - Distribute the computation on the Grid
- Benefit
 - Expect 10x faster simulation speed
 - Ability to model much larger problem within a reasonable time
- Part of TNGP/ Hewlett Packard Join Research Program



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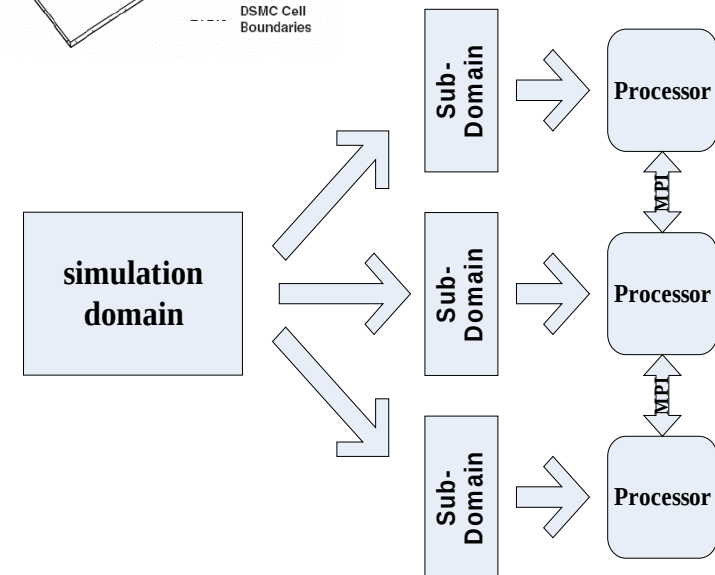
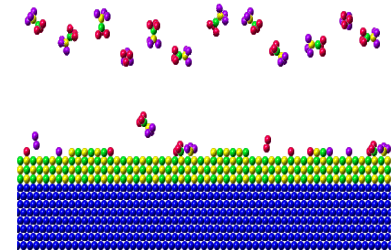
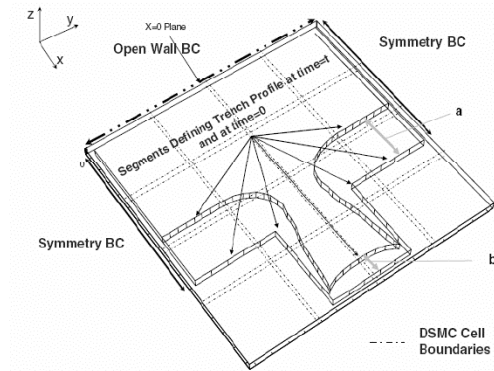
The Parallel Simulation Model for Thin Film Deposition Using the DSMC Method

Dr. Rardchawadee Silapunt
Dr. Tiranee Achalakul
KMUTT, Thailand

- Thin film deposition process is used for many manufacturing industries including very large scale integrated circuits, optics, magnetic, hard and corrosion resistant coatings, micro-mechanics.
- Grid system will be used to accelerate the computation and simulation process
- Part of TNGP/ Hewlett Packard Join Research Program

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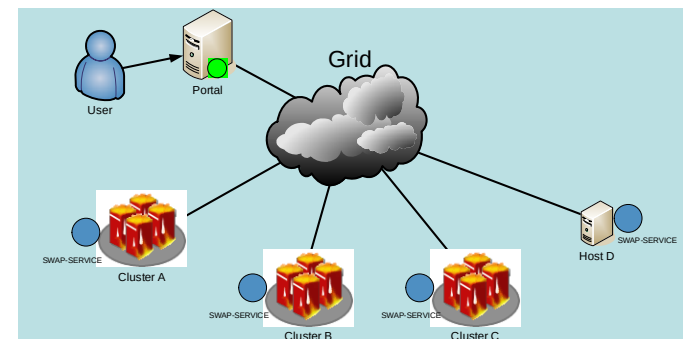
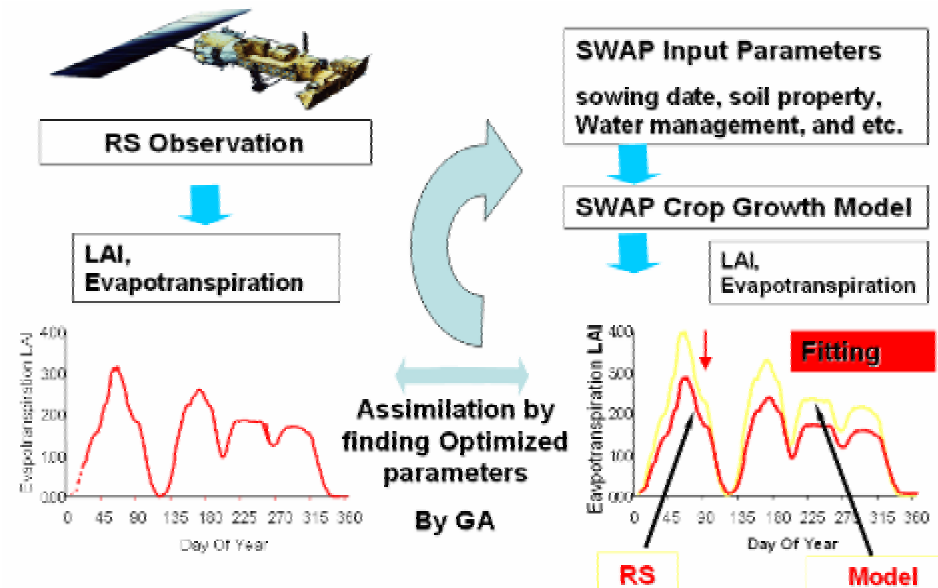
<http://www.thaigrid.net>



Grid based data assimilation using RS data



- Dr. Kiyoshi Honda, AIT
- Dr. Putchong Uthayopas, THNC
- Challenges
 - Using GA based data assimilation for agricultural monitoring
 - Very long computation time from months to years
 - Large amount of RS data needed to be moved around and process
- Solution:
 - Using Grid to harvest more computing power
 - Hide data assimilation process behind the service using MPI, GridRPC (ninf-g)
- Benefit
 - Speeding up the calculation



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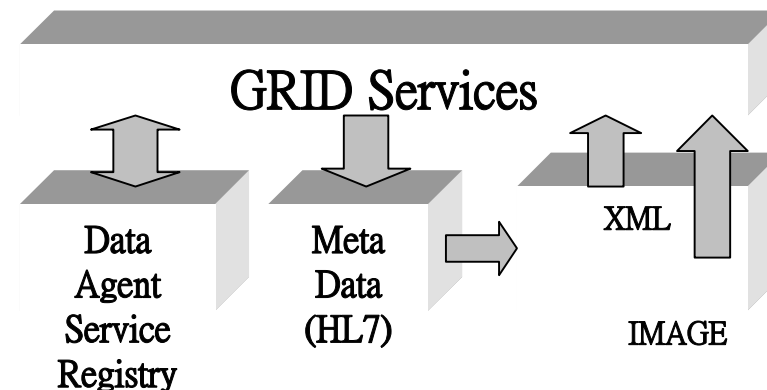
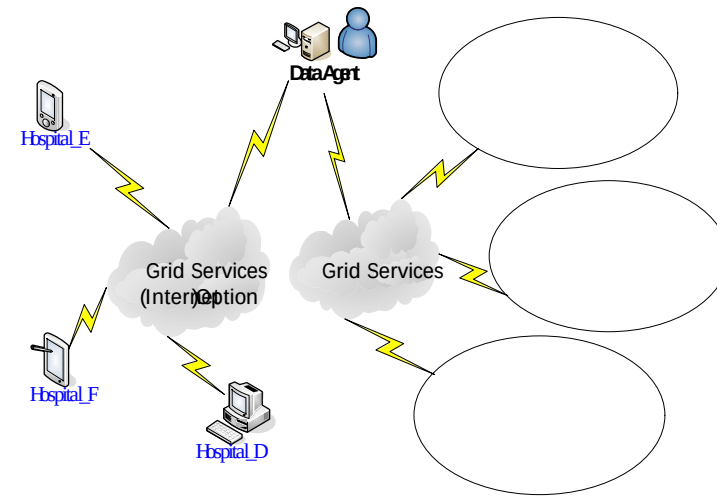
Medical Grid



Suwachai Siemoon, Somkiat
Chatchuenyot, Dr. Panjai
Tantasawong

Computer Science Department,
Silpakorn University

- To build and develop a distributed information management system using DataGrid technology.
- Apply the distributed information management system to test on exchanging and managing the Patient Records System, which is record complexity and, different format and large size of data.



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TNGP Role



- Provide Grid Nodes to founding members
- Provides access to TNGP facilities
 - Teraflop Grid Server
- Oversee the construction and reliable operation of the infrastructure
- Helps partners institute in application porting and grid adoption
- Conduct activities
 - Technology transfer and training
 - Grid technology promotion
 - International and national community building and participation



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Partners Institute Roles



- Defining project direction
- Participating in TNGP activities
- Promoting and stimulating grid technology adoption internally and externally
- Conduct research and development on Grid technology
- Dedicate necessary manpower, expertise, and resources to make the project successful
 - Support Grid Node



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Target



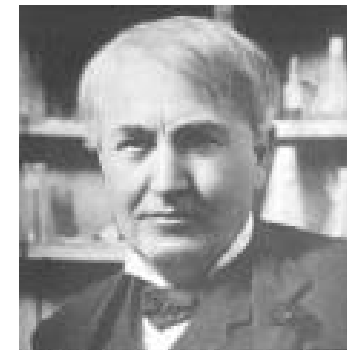
- Buildup 2,000 grid engineers in 3 years
- 10 industry showcases or more
- Community of grid researchers consists of 200-300 people
- About 20-30 in depth research projects
- Thailand participation in large international grid project
 - TeraGrid
 - PRAGMA
 - APAN



Thank You



Collaborations Welcome



panupong@kku.ac.th

